Eenheid 2.3

Note Title 2016/03/10

Limietwette

(Sien Clickup)

Stewart bl 95 – 104

Studiegids bl. 14

Bepaal indien moontlik

① $\lim\limits_{x \to 2} \left(3x - 5\cos \pi x + 4\sin \frac{\pi}{x}\right)$

$= \lim\limits_{x\to 2} 3x - \lim\limits_{x\to 2} 5\cos \pi x + \lim\limits_{x\to 2} 4\sin \frac{\pi}{x}$

$= 6 - (5)(1) + 4(1)$

$= 5$

② $\lim_{x \to 3} \frac{3-x}{x^2-9}$ $\left(\frac{0}{0}\right)$ Faktoriseer

$= \lim_{x \to 3} \frac{3-x}{(x-3)(x+3)} = \lim_{x \to 3} \frac{-\cancel{(x-3)}}{\cancel{(x-3)}(x+3)}$

$= -\frac{1}{6}$

③ $\lim_{x \to \pi/2} \sqrt{x} \times \sin x = \lim_{x \to \pi/2} \sqrt{x} \times \lim_{x \to \pi/2} \sin x$

$= \left(\sqrt{\pi/2}\right)(1) = \sqrt{\pi/2}$

④ $\lim_{x \to 1} \frac{3-x}{x^2-9} = \frac{2}{-8} = -\frac{1}{4}$

Toegevoegde

⑤ $\lim_{x \to 0} \frac{x}{\sqrt{x+4} - 2} \left(\frac{0}{0}\right)$ plan

$= \lim_{x \to 0} \frac{x}{\sqrt{x+4} - 2} \times \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2}$

$= \lim_{x \to 0} \frac{x(\sqrt{x+4} + 2)}{(x+4) - 4} = \lim_{x \to 0} \frac{\cancel{x}(\sqrt{x+4} + 2)}{\cancel{x}}$

$= \sqrt{4} + 2 = 4$

Knyptangstelling p 101

As $f(x) \leq g(x) \leq h(x)$ vir x naby a en $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a} h(x) = L$

dan is $\lim\limits_{x\to a} g(x) = L$.

Onthou: **Metodes om limiete te bepaal:**

- Direkte Substitusie
- Faktorisering
- Halwe limiete
- Limietwette
- Toegevoegde
- Knyptangstelling

Knyptangstelling meestal vir $\sin x$ en $\cos x$

- $-1 \leq \sin x \leq 1$
- $-1 \leq \cos x \leq 1$
- $-\frac{\pi}{2} < \text{bgtan}\, x < \frac{\pi}{2}$

Voorbeeld: Bepaal $\lim_{x\to 0^+} \sqrt{x}\sin\frac{1}{x}$

Opmerking: Kan nie limietwette gebruik nie want $\lim_{x\to 0^+} \sin\frac{1}{x}$ bestaan nie.

kleiner

$\lim_{x\to 0^+} \sin\frac{1}{x} = +\infty$ ~~

Plan: Knyptangstelling: $\lim_{x\to 0^+} \sqrt{x}\sin\frac{1}{x}$

Skep

$-1 \le \sin\frac{1}{x} \le 1$ as $x \neq 0$

$\Rightarrow (-1)\times\sqrt{x} \le \sqrt{x}\sin\frac{1}{x} \le (\sqrt{x})(1)$

Nou $\lim_{x\to 0^+} -\sqrt{x} = 0$ en $\lim_{x\to 0^+} \sqrt{x} = 0$

Dus $0 \le \lim_{x\to 0^+} \sqrt{x}\sin\frac{1}{x} \le 0$ Dan is $\lim_{x\to 0^+} \sqrt{x}\sin\frac{1}{x} = 0$

4 Spesiale limiete

(ken dit) asook toepassings

VASTE STELLINGS

- $\lim\limits_{x \to \text{"}0\text{"}} \dfrac{\sin\text{"}x\text{"}}{\text{"}x\text{"}} = 1$ antw

- $\lim\limits_{x \to \text{"}0\text{"}} \dfrac{\text{"}x\text{"}}{\sin\text{"}x\text{"}} = 1$ $\left(\begin{matrix} x \to 0 \\ \frac{0}{\sin 0} \end{matrix} \right)$

- $\lim\limits_{x \to \text{"}0\text{"}} \dfrac{\tan\text{"}x\text{"}}{\text{"}x\text{"}} = 1$

- $\lim\limits_{x \to \text{"}0\text{"}} \dfrac{\text{"}x\text{"}}{\tan\text{"}x\text{"}} = 1$

$$\boxed{\begin{matrix} \text{Dink } x \to 0 \\ \text{Sien } \dfrac{\sin 0}{0} \end{matrix}}$$

VOORBEELDE

① $\lim\limits_{x \to \pi} \dfrac{\sin \overset{\pi}{x}}{\underset{\pi}{x}} = \dfrac{0}{\pi} = 0$ Direkte subst.

(2) $\displaystyle\lim_{x \to 0} \frac{\sin 4x}{x}$ Plan $\times \frac{4}{4}$

$x \to 0$

$4x \to 0$

$= 4 \displaystyle\lim_{4x \to 0} \frac{\sin 4x}{4x} = 4 \times 1 = 4$

$\left(\text{Dint} \quad \frac{\sin 0}{0} \quad x \to 0 \right)$

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(3) $\displaystyle\lim_{x \to 2} \frac{\tan(x^2-4)}{x-2}$? ~ bly so

Plan

$x \to 2$

$\Rightarrow x-2 \to 0$

$= \displaystyle\lim_{x \to 2} \frac{\tan(x^2-4)}{x-2} \times \lim_{x \to 2} \frac{x+2}{x+2}$

$x^2-4 = (x-2)(x+2)$

$= \boxed{\displaystyle\lim_{x \to 2} \frac{\tan(x^2-4)}{x^2-4}} \times \lim_{x \to 2} (x+2)$

$= 1 \times 4 = 4$

4

$$\lim_{x \to 0} \frac{x^2}{\sin x} \rightarrow x \times x$$

$$\lim_{x \to 0} \frac{x}{\sin x} = 1$$

$$\lim_{x \to 0} \frac{0}{\sin 0}$$

$$= \lim_{x \to 0} \frac{x}{\sin x} \times \lim_{x \to 0} x$$

$$= 1 \times 0 = 0$$

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$$\lim_{x \to 0} \frac{\tan x^2}{x} \times \frac{x}{x} \; ?$$

$$= \boxed{\lim_{x \to 0} \frac{\tan x^2}{x^2}} \times \lim_{x \to 0} x$$

$$= 1 \times 0$$

$$= 0$$

6) $$\lim_{x\to 0} \frac{\tan 6x}{\sin 2x} = \lim_{x\to 0} \tan 6x \times \lim_{x\to 0} \frac{1}{\sin 2x}$$

$$= \lim_{x\to 0} \tan 6x \times \frac{6x}{6x} \times \lim_{x\to 0} \frac{1}{\sin 2x} \times \frac{2x}{2x}$$

$$= \boxed{\lim_{x\to 0} \frac{\tan 6x}{6x}} \times \boxed{\lim_{x\to 0} \frac{2x}{\sin 2x}} \times \lim_{x\to 0} \frac{6\cancel{x}}{2\cancel{x}} \text{?}$$

Vereenvoudig

$$= 1 \times 1 \times \frac{6}{2}$$

$$= 3$$

Metodes om limiete te bepaal

OPSOMMING

Tegnieke

① Direkte substitusie

$$\lim_{x \to \pi} \frac{\tan x}{x} = \frac{0}{\pi} = 0$$

② Vorm $\left(\frac{0}{0}\right)$: Faktoriseer

$$\lim_{t \to -3} \frac{t^2 - 9}{2t^2 + 7t + 3} \left(\frac{0}{0}\right) = \lim_{t \to -3} \frac{(t-3)\cancel{(t+3)}}{(2t+1)\cancel{(t+3)}}$$

$$= \lim_{t \to -3} \frac{t-3}{2t+1}$$

$$= \frac{-6}{-5} = \frac{6}{5}$$

③ Oneindige limiete

$$\lim_{x\to 0^-}\left(\frac{1}{x}-\frac{1}{|x|}\right)$$

$\downarrow$ beteken $x<0 \therefore |x| = -x$

Gebruik definisie van $y=|x|$

$$|x| = \begin{cases} x & \text{as } x>0 \\ -x & \text{as } x<0 \end{cases}$$

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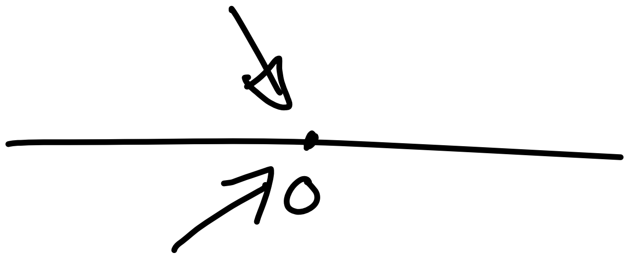


$$= \lim_{x\to 0^-}\left(\frac{1}{x}-\frac{1}{(-x)}\right)$$

$$= \lim_{x\to 0^-} \frac{\overset{\text{pos}}{2} \neq 0}{x \rightsquigarrow}$$

al hoe kleiner neg

$$= \boxed{-}\infty$$

④ Limietwette (op ClickUP)

⑤ <u>Toegevoegde</u>

$$\lim_{h \to 0} \frac{\sqrt{9+h} - 3}{h} \quad \left(\frac{0}{0}\right)$$ ⟶ onbepaalde vorm

$$= \lim_{h \to 0} \frac{\sqrt{9+h} - 3}{h} \times \frac{\sqrt{9+h} + 3}{\sqrt{9+h} + 3}$$

(vereenvoudig)

$$= \lim_{h \to 0} \frac{9+h-9}{h(\sqrt{9+h}+3)}$$

$$= \lim_{h \to 0} \frac{\cancel{h}}{\cancel{h}(\sqrt{9+h}+3)}$$

$$= \frac{1}{6}$$

(6) Knyptangstelling

Bepaal: $\lim_{x \to 0} x^4 \cos \frac{2}{x}$

as $x \neq 0$ dan

$-1 \leq \cos \frac{2}{x} \leq 1$

$\Rightarrow (-1)x^4 \leq x^4 \cos \frac{2}{x} \leq x^4$

Nou $\lim_{x \to 0} -x^4 = 0$

en $\lim_{x \to 0} x^4 = 0$

Dan is $\lim_{x \to 0} x^4 \cos \frac{2}{x} = 0$ volgens

Knyptangstelling

(7) "Vaste Stellings"

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \to 0} \frac{\sin(x-2)}{x^2-4}$$

$$x^2 - 4 = (x-2)(x+2)$$

$$= \boxed{\lim_{x \to 0} \frac{\sin(x-2)}{(x-2)}} \times \lim_{x \to 0} \frac{1}{x+2}$$

$$= 1 \times \frac{1}{2}$$

$$= \frac{1}{2}$$

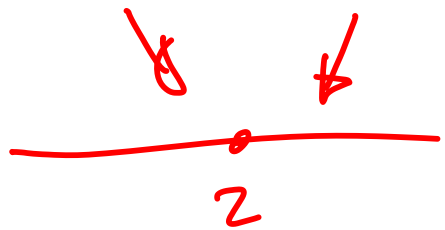
① $\lim\limits_{x\to \pi/2} \dfrac{\cos 2x}{x} = \dfrac{-2}{\pi}$

Jien: $\dfrac{\cos \cancel{2}(\pi/\cancel{2})}{\pi/2} = \dfrac{\cos\pi}{\pi/2} = \dfrac{-1}{\pi/2}$

② $\lim\limits_{x\to 3} \dfrac{x-3}{\underbrace{x^2-2x-3}_{(x+1)(x-3)}} \left(\dfrac{0}{0}\right) = \lim\limits_{x\to 3} \dfrac{1}{x+1} = \dfrac{1}{4}$

③ $\lim\limits_{x\to 2} \dfrac{x^2-x+6}{x-2}$ $\rightsquigarrow 8 \neq 0$

$\rightsquigarrow$ kleiner



$\lim\limits_{x\to 2^+} \dfrac{x^2-x-6}{x-2}$ pos $\neq 0$

$\rightsquigarrow$ klein pos

$= +\infty$

$$\lim_{x \to 2^-} \frac{x^2 - x - 6}{x - 2} = -\infty$$

pos $\neq 0$

$\rightsquigarrow$ kleiner neg

En $\lim_{x \to 2} f(x)$ bestaan nie want

$\lim_{x \to 2^-} f(x) = -\infty$ en

$\lim_{x \to 2^+} f(x) = \infty$

VA by $x = 2$!

- $\lim_{x \to 0^+} \sqrt{x} \operatorname{bgtan} \frac{1}{x}$